

MICHAL HORNICKY



SENIOR SOFTWARE ENGINEER / AGENT INFRASTRUCTURE / ML SYSTEMS

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00 / CURRENT FOCUS

Self-modifying agentic environments. RLVR environments and the data they generate. Constitutional AI for agent systems. The environment side of recursive self-improvement: the harness, the tooling, the feedback loops around the model.

01 / ABOUT

- Engineer at **Google DeepMind** on the Gemini team; ten years across ML systems, distributed infrastructure, and large-scale applications; in my free time, a fleet of my own agent systems
- I work **prototype-first**: try the idea, survey the state of the art, then turn the exploration into a design doc and a shipped system
- I optimize for **developer velocity**: every other metric can be improved if velocity is fast
- Alignment-aware**: the systems I build ensure alignment through the environment and reduce hallucinations, model drift, and reward hacking

02 / EXPERIENCE

Senior Software Engineer · Gemini / Google DeepMind

Nov 2024 – Present

Google · Zürich

- Technical lead of a small team working on **agent infrastructure and product quality** for the Gemini application: agent development environments where fleets of collaborating agents take on planning, implementation, and review
- Early adopter of agents across the Gemini application stack; built many of the **early integration systems**: multi-agent orchestration, tool use, and the internal best practices around them
- Developed an **agent-driven testing and QA framework** from scratch; the team is now building new capabilities on top of it
- Drive adoption through tooling and workshops; the job is researcher and engineer **velocity**, making the people around me faster with better machinery
- Implemented product features in the Gemini mobile application

Software Engineer · Assistant & Gemini

Nov 2021 – Nov 2024

Google · Zürich

- Developed automated test infrastructure and tooling for Android and iOS applications
- Supported testing and releases for Google Assistant, Search, and Gemini/Bard mobile apps
- Built early internal LLM tools for automated testing and software verification

Senior Software Engineer / Team Lead

Jul 2020 – Sep 2021

Red Hat · Brno

- Led a team of 5 engineers building the Patch application on Red Hat Insights (cloud.redhat.com)
- Engineered a custom in-memory **dependency solver** and evaluation engine for RHEL package resolution after PostgreSQL hit its limits on graph traversal
- Scaled the platform to serve thousands of RHEL fleet deployments daily

Software Engineer

Sep 2019 – Jul 2020

Red Hat · Brno

- Initiated development of the Patch application; built Go libraries for integrating into the Python ecosystem

Android Developer

Mar 2016 – Sep 2019

TouchArt s.r.o · Brno

- Full lifecycle management of 10+ B2B Android applications; led architectural modernization to Kotlin, MVVM, reactive dataflow

Bedrock · Autonomous Software Engineering Platformsemtexzv.github.io/bedrock

A fleet of coding agents that builds, reviews, and ships a software platform end to end · 2026 – Present

- A **control plane for coding agents**: a fleet of ≈15 collaborating agents that build, review, and integrate changes into a shared monorepo, end to end
- Built on **Buck2** and **jj**: a self-integrating monorepo with hermetic builds, CI gating, and a web review front end, so every change an agent makes is reproducible and lands through one pipeline
- Generates its own data**: every change, dispatch, conversation, and review is a logged trajectory. The direction is the platform as an **RL environment**, with rubric-scored trajectories (**rubrics as rewards**) and **GRPO-style post-training** of its own agents on its own data
- Bounded self-modification, designed against reward hacking**: a structural security layer protects the core, changes to it require human approval, and the automated reviewer is advisory, never the sole authority on an adversarial change. Safety is a structural property, not a policy

llm-lab · Exploratory LLM Traininggithub.com/semtexzv/llm-lab

A home-run LLM architecture research platform, executing an autoresearch loop targeting small models · 2025 – Present

- Verifies or falsifies my hypotheses about LLM architecture**, with the goal of finding **more efficient architectures** that reduce training costs; results are analyzed on **FLOPs, memory, bandwidth, and token efficiency**
- Targets **Apple MLX** and **PyTorch/CUDA** for comparative performance analysis, validated against a measured hardware bandwidth model
- Currently focused on **novel architectures of the residual stream** in decoder-only transformers, inspired by manifold hyper-connections ([arXiv:2512.24880](https://arxiv.org/abs/2512.24880)) and attention residuals ([arXiv:2603.15031](https://arxiv.org/abs/2603.15031))

mage · Self-Recompiling Coding Agentgithub.com/semtexzv/mage

An agent that extends itself by recompiling its own binary

- A **Darwin Gödel Machine-style self-modifying agent** at the compiled-artifact level: embeds a snapshot of its own source, splices in new modules, recompiles a fresh binary, and hot-swaps into the next generation under a supervising monitor
- Rust workspace (edition 2024): concurrent/serial tool dispatch, hierarchical cancellation, and a differential terminal renderer

Termineer · Autonomous AI Coding Agentgithub.com/semtexzv/termineer

Self-directed coding agent built on a configurable tool-calling grammar

- A **provider-agnostic function-calling abstraction**: one configurable grammar unified tool calling across Anthropic, Google, and OpenAI, built when providers had no shared format for it
- Autonomous task decomposition with specialized sub-agents; MCP protocol support

GenAgent · Always-On Personal Assistantgithub.com/semtexzv/genagent

Taskbar AI assistant for macOS/iOS, explored alongside Termineer

- Menu-bar assistant invocable anytime for coding and general help

Rube · Generalized Control-Loop Architecturegithub.com/semtexzv/rube

- A generalized reconciliation and control-loop runtime: the Kubernetes **operator pattern** applied to arbitrary resources, desired-state management without the YAML or the containers

Protokit · High-Performance Protocol Buffers for Rustgithub.com/semtexzv/protokit

- Only Rust implementation supporting the full TextFormat specification; competitive with **prost**; dual front-end compiler

Stor · Embedded Key-Value Databasegithub.com/semtexzv/stor

- Typed key-value database on RocksDB with zero-copy deserialization; storage engine for a low-latency portfolio platform

05 / SKILLS	AGENT SYSTEMS	Autonomous coding agents , Multi-agent orchestration, Agent harness design, RL environments, Context engineering, Tool calling, MCP protocol
	ML / AI	PyTorch , MLX , Post-training (SFT, RLVR, GRPO), Evals, Training pipelines, Distributed training, Reinforcement learning, Transformer architectures, Autodiff
	SYSTEMS	Distributed systems, HPC, GPGPU (CUDA / OpenCL), ML accelerators, Hermetic builds, Sandboxing, Actor models
	INFRASTRUCTURE	Kubernetes, Terraform, Large-scale data processing, Observability, CI / CD, GCP
	LANGUAGES	Python , Rust , Go, C / C++, Swift, Kotlin, TypeScript
	SPOKEN	Slovak (Native), Czech (Native), English (Fluent), German (Beginner)

06 / EDUCATION	Master's Degree, Information Technology	2017 – 2019
	Brno University of Technology · Brno —Thesis: distributed algorithmic trading system (Rust, ZeroMQ, NATS, Kubernetes)	
	Bachelor's Degree, Information Technology	2014 – 2017
	Brno University of Technology · Brno	